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Macroscale Gradient Dysfunction in Alzheimer's Disease: Patterns With Cognition Terms and Gene Expression Profiles

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ABSTRACT

Macroscale functional gradient techniques provide a continuous coordinate system that extends from unimodal regions to transmodal higher-order networks. However, the alterations of these functional gradients in AD and their correlations with cognitive terms and gene expression profiles remain to be established. In the present study, we directly studied the functional gradients with functional MRI data from seven scanners. We adopted data-driven meta-analytic techniques to unveil AD-associated changes in the functional gradients. The principal primary-to-transmodal gradient was suppressed in AD. Compared to NCs, AD patients exhibited global connectome gradient alterations, including reduced gradient range and gradient variation, increased gradient scores in the somatomotor, ventral attention, and frontoparietal regions, and decreased in the default mode network. More importantly, the Gene Ontology terms of biological processes were significantly enriched in the potassium ion transport and protein-containing complex remodeling. Our compelling evidence provides a new perspective in understanding the connectome alterations in AD.

Dawei Wang and Zhuangzhuang Li contributed equally to this work.

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1 | Introduction

Alzheimer's disease (AD) is characterized by a progressive deterioration in primary and higher-order cognitive functions and memory, which can disturb daily living ability, eventually resulting in dementia (Jia, Quan, et al. 2020). AD typically destroys neurons and their connections in parts of the brain involved in memory, including the entorhinal cortex and hippocampus. We sometimes notice older adults moving slower or having difficulty navigating our environment with disrupted hearing and lower visual and sensing abilities. Over time, a person with AD gradually loses their primary ability to live and function independently. It then affects areas in the cerebral cortex responsible for higher-order cognition, such as language, reasoning, and social behavior (Jia, Du, et al. 2020; Jia, Quan, et al. 2020). Large-scale brain networks provide a coherent framework for understanding cognition by offering a neural model of how different cognitive functions are impaired in AD. Herein, an empirical description of brain network function is needed to guide the understanding of the primary human function as a biological foundation for clarifying the disrupted connectivity patterns of AD (Bernhardt et al. 2022; Hu et al. 2022).

Convergence evidence based on connectome studies has demonstrated that hierarchy is an optimal organization for information transferring within the human brain (Dong, Margulies, et al. 2021; Hu et al. 2022; Tian et al. 2020; Xia, Xia, et al. 2022). Traditional resting-state functional magnetic resonance imaging (rs-fMRI) is the most widely used technique to study the changes in functional activity in brain networks (Bernhardt et al. 2022; Raut, Snyder, and Raichle 2020; Smith et al. 2012). Interestingly, spontaneous slow fluctuations in spatial district brain regions showed hierarchical dynamics organization (Pines et al. 2023). Advances in neuroimaging technologies and analytics provide a continuous coordinate system enabling us to capture the cerebral cortex's gradient topological organization, which helps reveal a principal primary-to-transmodal gradient in the brain network (Dong, Yao, et al. 2021; Margulies et al. 2016; Paquola et al. 2019; Tian et al. 2020; Vogel et al. 2020; Xia, Xia, et al. 2022). A connectome gradient reflects systematic variations in brain connectivity patterns along a continuous spectrum. This gradient exhibits a spatially continuous transition in functional connectivity, progressing from unimodal (sensory-motor) regions to transmodal (default mode) regions (Margulies et al. 2016). This pattern is aligned with the hierarchical organization of the brain. Notably, these transmodal regions have been implicated in various diseases, demonstrating alterations at structural, functional, and molecular levels (Xue et al. 2023).

AD pathology profoundly affects functional connectomes (Corriveau-Lecavalier et al. 2023; Xie and He 2011), and changes initially occur from primary daily, temporal, and attention functions. Then the alterations are followed by dysfunction and connectivity remodeling between primary and high-order brain network areas (Yu, Sporns, and Saykin 2021). Network tools are precious for studying AD, characterized by the stereotyped spatial spread of disruption along different neural pathways. Therefore, the study of macroscale functional gradient techniques is of particular importance in detecting the key characteristics of how functional domains are reorganized from unimodal (sensory visual and motor networks) regions to the transmodal networks (default mode network, [DMN]) (Xia, Liu, et al. 2022). Elaborating on characterizing the principal

connectome gradient in AD could provide new insight into the hierarchical network mechanisms for understanding the concurrence of unimodal-transmodal dysfunctions and offer potential biomarkers for AD (Perovnik et al. 2023). Recently, based on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, gray matter network hierarchy differences were explored in individual MCI patients who converted to AD and nonconverters. The findings revealed a more compressed gray matter network hierarchy in patients who converted to AD (Zhou, Zhao, and Wu 2024). Another study identified a decreased gradient in the default mode and frontoparietal networks in AD patients (Zheng et al. 2024). However, both studies were based on public datasets with relatively small sample sizes. Even though a recent study suggested AD-associated brain network hierarchy reorganization (Hu et al. 2022), how the network hierarchy changes of gradients in microstructure network, connectivity, gene expression, and function in AD remains unclear.

Regarding AD, it has been well-evaluated that neurodegeneration disruption is modulated by genetic factors (Nativio et al. 2020; Scelsi et al. 2018; Tan et al. 2019; Thijssen et al. 2020). Recently, transcriptome-connectome association studies (Hansen et al. 2021; Seidlitz et al. 2020, 2018; Sydnor et al. 2021) have provided a critical opportunity to bridge the gap between the microscale transcriptome profile and the macroscale brain network. Specifically, the functional architectures of the connectomes are associated with gene expression profiles involving ion channel activity and oxidative metabolism (Richiardi et al. 2015). Therefore, we speculate that AD patients exhibit transcriptome profile-regulated disturbances in the primary to high-order connectome gradient. The elucidation of the relationship between gradient and transcriptome profile can enhance our understanding of the molecular genetic underpinnings of the dysfunction in AD.

This study aims to investigate the alterations of functional gradients in AD and their associations with cognitive terms and gene expression profiles. Firstly, we analyzed the functional gradients in a biobank with 809 samples from seven neuroimaging centers. We introduced meta-analytic techniques to unveil reliable changes in the connectome gradients in AD. Secondly, we investigated the cortex-wide gradient changes with a region of interest (ROI)-wise comparisons to identify the abnormal brain unimodal or transmodal areas. Specifically, we also investigated whether the AD-associated impaired connectome gradients linked to gene expression profiles enriched biological processes.

2 | Material and Methods

2.1 | Subjects

This study included 813 subjects scanned on 3.0T MRI scanners and recruited from four hospitals in China. All subjects had demographic and clinical information. Demographics and clinical characteristics of datasets are shown in Table 1. The multi-center database included 296 AD patients (age: 68.89 ± 8.28 , gender (M/F): 134/162; MMSE: 16.56 ± 6.03 ; MOCA: 13.49 ± 4.91), 258 MCI patients (age: 68.55 ± 8.92 , gender (M/F): 125/133; MMSE: 25.13 ± 3.39 , MOCA: 20.28 ± 3.83), and 259 NCs (age: 66.93 ± 6.84 , gender (M/F): 105/152; MMSE: 28.52 ± 1.65 , MOCA: 26.33 ± 2.33). It should be noted that the

TABLE 1 | Demographics and clinical characteristics of datasets.

	NC (N=259)	MCI (N=258)	AD (N=296)	<i>p</i> (CN vs. MCI vs. AD)	<i>p</i> (CN vs. MCI)	<i>p</i> (CN vs. AD)	<i>p</i> (MCI vs. AD)
Age	66.93 ± 6.84	68.55 ± 8.92	68.89 ± 8.28	0.01	0.02	0.02	0.64
Gender (M/F)	105/152	125/133	134/162	0.65	0.42	0.84	0.58
MMSE	28.52 ± 1.64	25.14 ± 3.39	16.56 ± 6.03	<0.001	<0.001	<0.001	<0.001
CDR	0.00 ± 0.00	0.50 ± 0.00	1.31 ± 1.25	<0.001	<0.001	<0.001	<0.001
Delay recall	6.99 ± 2.99	3.20 ± 2.43	0.72 ± 1.16	<0.001	<0.001	<0.001	<0.001
MOCA	26.32 ± 2.33	20.27 ± 3.83	13.49 ± 4.91	<0.001	<0.001	<0.001	<0.001

diagnosis of AD was primarily based on clinical evaluation followed by NINCDS-ADRCA criteria (McKhann et al. 1984). All participants have provided written informed consent. Briefly, the CN participants were required to meet the following inclusion criteria: (1) no memory complaints; (2) clinical dementia rating=0. Inclusion criteria for participants with MCI were as follows: (1) participants must express a subjective memory concern; (2) CDR=0.5; Inclusion criteria for patients with AD were as follows: (1) participants must express a subjective memory concern; (2) CDR=1.0 or 2.0. Detailed subject information is provided in S01 and found elsewhere in our previous studies (Chen et al. 2023; Du et al. 2022).

2.2 | Imaging Acquisition and Preprocessing

For each sample, resting-state fMRI data were obtained using 3.0-T MRI scanners (S01) and preprocessed using the BRAINetome Toolkit (<http://brant.brainnetome.org/>) (Xu et al. 2018) with a standard pipeline including the following steps: (1) slice timing correction; (2) realignment to the first volume; (3) spatial normalization to MNI space at 2×2×2 mm; (4) regression of nuisance signals, including linear trends, six motion parameters, and their first-order differences, and signals representing white matter and cerebrospinal fluid; and (5) temporal bandpass filtering (0.01–0.08 Hz) to reduce high-frequency noise. Detailed information about data preprocessing is described in our previous studies (Chen et al. 2023; Jin, Wang, et al. 2020).

2.3 | Statistical Analysis

All statistical analyses were performed using Python. The chi-squared test was used for the gender comparisons. The analysis of variance (ANOVA) was used for the age and MMSE comparisons between NC, MCI, and AD groups. A *p* value < 0.05 was considered statistically significant.

2.4 | Macroscale Connectome Gradient Analysis

The gradient analysis was implemented using the BrainSpace toolbox (<https://github.com/MICA-MNI/BrainSpace>). We parcellated the cortex into 210 regions based on the Brainnetome atlas for the subsequent gradient analysis. For each subject, a functional (FC) matrix (210×210) was generated by calculating

the Pearson correlations between each pair of ROIs. We then performed Fisher z-transformation to improve the FC values' normal distribution. According to the connected weight, the top 10% of the connections were retained to keep sparsity, and other elements were set to 0 (Dong, Margulies, et al. 2021). After that, we calculated the cosine similarity matrix between each pair of ROIs and scaled the similarity matrix into a normalized angle matrix. Next, the diffusion map embedding method was applied to identify the principal gradient components explaining the connectome variance in descending order. The diffusion map embedding dimension-reduction algorithm estimates a low-dimensional manifold from a high-dimensional FC matrix and is robust with noise (Margulies et al. 2016). Finally, the resulting embedding as a gradient score was applied to align components of each individual to the brain template using the Procrustes rotation method (Vos de Wael et al. 2020).

Next, we used a linear regression model to reduce the confounding effects of age and gender within each site. After that, a meta-analytic framework was performed on the gradient score using the Python Meta-Analysis library (<https://github.com/YongLiulab/>) (Details can be found in S02). A random-effects model weighed the sites before aggregating the effect size (Higgins 2009). For effect size estimates, we used Cohen's *d* value.

$$\text{Cohen's } d = \frac{\bar{x}_1 - \bar{x}_2}{s} \quad (1)$$

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (2)$$

where $\bar{x}_1, \bar{x}_2$ are the gradient means, n_1, n_2 are the sample sizes, s_1, s_2 are the standard deviations of the two groups. The differences in the gradient scores between the NCs and AD groups were quantitatively analyzed using the meta-analysis method. The z-statistic was obtained as follows:

$$z = \frac{\sum_i \text{cohen's } d_i \times \omega_i}{\sqrt{\sum_{i=1} \omega_i}} \quad (3)$$

where Cohen's d_i represents the effect size of the *i*-th site and ω_i is the within-site variance between-sites variance of *i*-th site (*i* = 1, ..., 7 in the present study) plus between-sites variance. For a two-tailed test, the *p* value was given by.

$$p = 2(1 - \varphi(z)) \quad (4)$$

We introduced the Family-wise error rate correction to correct for multiple comparisons across the brain regions ($p < 0.05/N$, $N = 210$).

2.5 | Association Between Connectome Gradients and Topological Properties of Functional Brain Network in AD

Given that the development of the connectome gradient might be mediated by the integration and segregation of functional networks, as suggested by Xia, Liu, et al. (2022), we next examined whether and how changes in connectome gradients are associated with the topological changes of the functional networks in AD (Bellenguez et al. 2022). To this end, we used the BRAinNetome Toolkit (<https://github.com/YongLiuLab/brant>) (Xu et al. 2018) to compute the characteristic shortest path length (L_p) and the clustering coefficient (C_p) of each subject in the binary whole-brain network. C_p can quantify the segregation, while L_p reflects the integration of the functional network (Bullmore and Sporns 2009). An ANOVA analysis assessed the differences between groups in L_p and C_p after controlling the age and sex. Using Pearson's correlation, we investigated the association between the global gradient metrics and functional network topological measures. The results were expressed about a standard statistical threshold of $p < 0.05$.

2.6 | Relationship Between AD-Associated Connectome Gradient Changes and Meta-Analytic Cognitive Terms

Neurosynth (<https://neurosynth.org/>) is a platform for the automated synthesis of fMRI data (Yarkoni et al. 2011). Inspired by Xia, Liu, et al. (2022), we introduced the Neurosynth to investigate whether AD-associated connectome gradient alterations parallel meta-analytic cognitive terms. First, the thresholded z -map of the principal gradient group was divided into AD-positive (AD > NCs) and AD-negative maps (AD < NCs). Then, the correlations between meta-analytic cognitive terms and resultant maps were estimated in the database using the “decoder” function in Neurosynth. To test the statistical significance of the correlations, we conducted the spatial permutation tests (1000 times) by randomly shuffling across ROIs. Next, we removed the noncognitive terms and selected the top 30 cognitive representations (Xia, Liu, et al. 2022). Finally, the cognitive terms associated with the AD-related gradient alterations were plotted in a word cloud. The font size was scaled according to the correlation coefficients of meta-analytic cognitive terms and AD-related connectome gradient alterations.

2.7 | Association Between Regional Gradients and Clinical Measures

To explore the relationship between the altered regional gradient and clinical measures, the Pearson correlation coefficients

between MMSE scores and the altered regional gradient were calculated ($p < 0.05$).

2.8 | Relationship Between Alterations in Connectome Gradient in AD and Gene Expression Profiles

To test the second biological hypothesis, that is, the brain area with significant AD-related alterations in connectome gradient link with high levels of gene risk terms. The gene expression data was obtained from six NCs from the Allen Human Brain atlas (<http://human.brain-map.org/>) and was projected to the Brainnetome atlas (Table S3) using the Abagen toolbox (<https://abagen.readthedocs.io/en/stable/>), resulting in a $202 \times 15,633$ gene expression matrix (8 of 210 regions did not identified related genes in Abagen toolbox). The case-control gradient difference (z -value) was used as the independent variable, and the gene expression profiles were used as the dependent variable in a partial least squares (PLS) regression analysis (Zhao et al. 2022). The PLS components represent the linear combination of the weighted 15,633 gene expression scores. To test the statistical significance of the correlations, we used the spatial permutation approach described above to generate 5000 permutations of the data (Xia, Liu, et al. 2022). We then ranked all the genes according to the weights of PLS components. Finally, the gene set enrichment analysis was utilized to explore the AD-related functional significance. The Gene Ontology terms (GO terms) were introduced using the WEB-based gene set analysis toolkit for gene set enrichment (<http://www.webgestalt.org/>) at $p < 0.05$ (FDR corrected).

2.9 | Validation Analyses

To validate the reproducibility of AD-associated abnormal gradient patterns, we introduced the Willard atlas ($N = 433$) (Altmann et al. 2015) and Schaefer atlas ($N = 988$) (Schaefer et al. 2018) to reproduce the gradient patterns with the same pipelines. We mapped the brain regions into surface vertexes and then computed the correlations of functional gradient between different brain atlas at the vertex level. In addition, we also explored the robustness of AD gradient patterns by different sparsity strategies of functional connection matrix (5%, 10%, 15%) and with or without framewise displacement. More details are provided in S08.

3 | Results

3.1 | Principal Gradients in the AD, MCI, and NC Groups

The principle gradient results showed a gradual axis of connectivity variations that was organized by low-level sensory systems such as the primary and unimodal visual, somatosensory/motor, and auditory regions on the one end and the transmodal regions, mainly including the DMN on the other end (Figure 1A). The results are consistent with previous studies (Abrigo et al. 2019; Alderson-Day et al. 2016; Xia, Liu, et al. 2022; Xu et al. 2022). The principal and secondary gradients explained 33.17%, 33.07%, and 33.25% of the whole-brain connectome variance

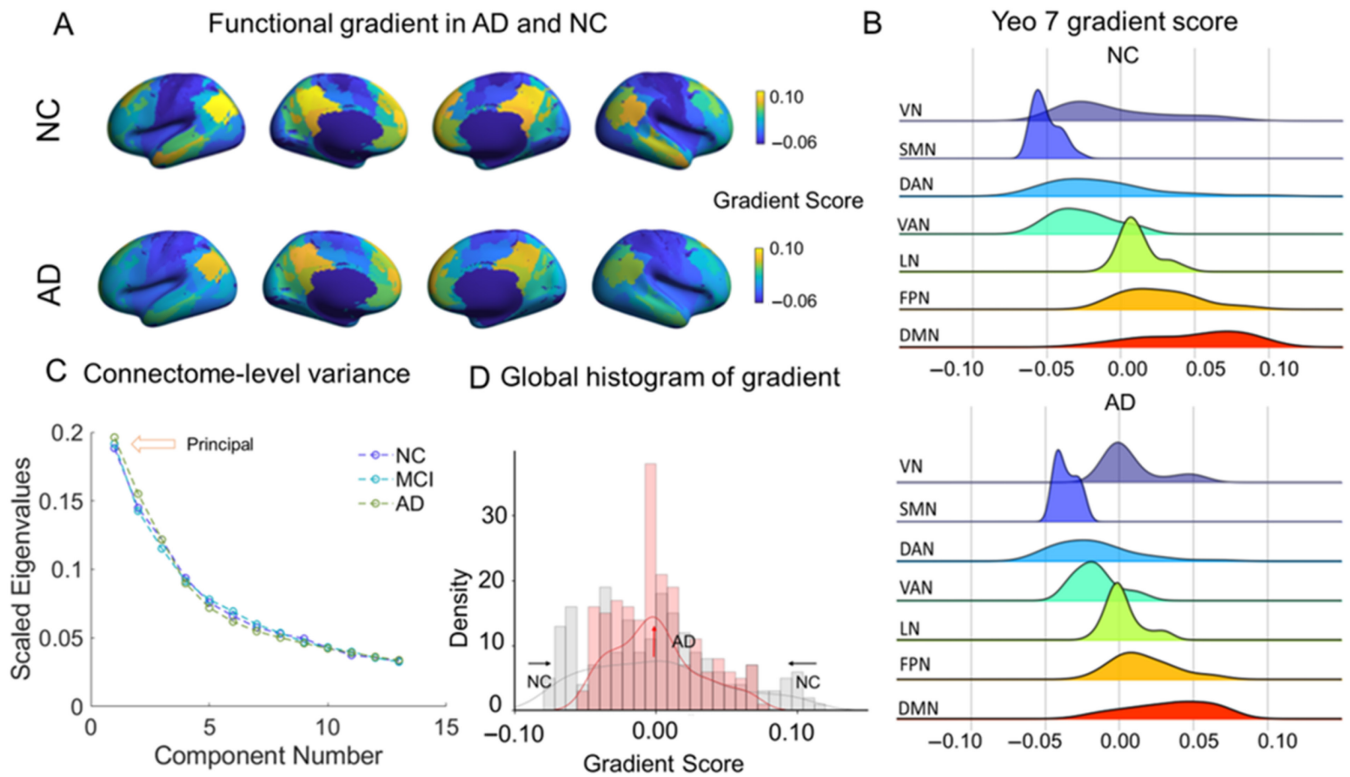


FIGURE 1 | (A) The principal gradient of connectivity in AD and NC groups. (B) A well-established functional community decomposition is applied to summarize surface-wide findings; the Brainnetome website provided the maps of the region of interest (ROIs) into the Yeo 7 networks (<http://atlas.brainnetome.org/download.html>). (C) The explanation ratio of connectome-level variance is explained by the top 13 components obtained from the diffusion embedding algorithm in AD, MCI and NC. (D) Global histogram of gradients in AD and NC groups. Abbreviations: DAN, dorsal attention network; DMN, default mode network; FPN, frontoparietal network; LN, limbic network; SMN, sensorimotor network; VAN, ventral attention network; VN, visual network.

in the AD, MCI, and NC groups, respectively (Figure 1C). We focused on the first two gradients, which account for a greater part of connectome variance and their association with neuronal microstructure (Changeux, Goulas, and Hilgetag 2021). Following the alignment of subject-wise gradients to a group-wise template, we compared gradient scores between AD and NCs after controlling the effects of gender and age. Globally the cortex-wide gradient was suppressed in AD. Both extremes (i.e., the somatomotor networks and DMN) were contracted relative to the control range, while primary gradients in the central axis increased in AD (Figure 1B,D). The result showed that AD has less spatial variation than NC.

The between-group comparisons showed that the primary-to-transmodal gradient explained less variance in the functional connectome in AD than those of MCI and NC groups ($F=10.34$, $p<0.001$, Figure 2A). Moreover, the AD patients showed a narrower range of scores than the MCI and NC groups ($F=9.32$, $p<0.001$, Figure 2A). Compared with NCs, AD showed increased gradient scores in the somatomotor, ventral attention, and frontoparietal regions, such as the superior temporal gyrus, middle frontal gyrus, and precentral gyrus, together with decreased gradient scores in the DMN regions and middle temporal gyrus ($p<0.05/N$, $N=210$, FWE corrected, Figure 2B). The altered gradient scores of AD capture the AD-specific hierarchical dysfunctions. Compared with MCI, the AD group showed a decreased gradient in the middle temporal gyrus and an increased

gradient in the insular gyrus ($p<0.05/N$, $N=210$, FWE corrected, Figure S8). There was no significant difference in gradient values between MCI and NCs at $p<0.05$ (FWE corrected).

3.2 | Correlation Between Connectome Gradients and Network Topology in AD

The patients with AD exhibited a lower clustering coefficient (C_p) than the MCI and NC groups ($F=3.91$, $p=0.02$), indicating disrupted segregation of AD brain networks. Moreover, C_p was significantly correlated with the global gradient range of the principal gradient across AD patients ($r=0.84$, $p<0.001$, Figure 2C). The gradient of AD was irrelevant to the shortest path length. The patients with AD exhibited a lower modularity than that in the NCs, which indicates disrupted segregation of brain networks in AD. Moreover, modularity was significantly correlated with the global gradient range of the principal gradient across AD patients. The gradient of AD was irrelevant to network efficiency (Figure S10).

3.3 | Meta-Analytic Cognitive Functions Related to Gradient Changes in AD

The associations between the meta-analytic map of cognitive terms and AD-related alterations in the primary gradient are

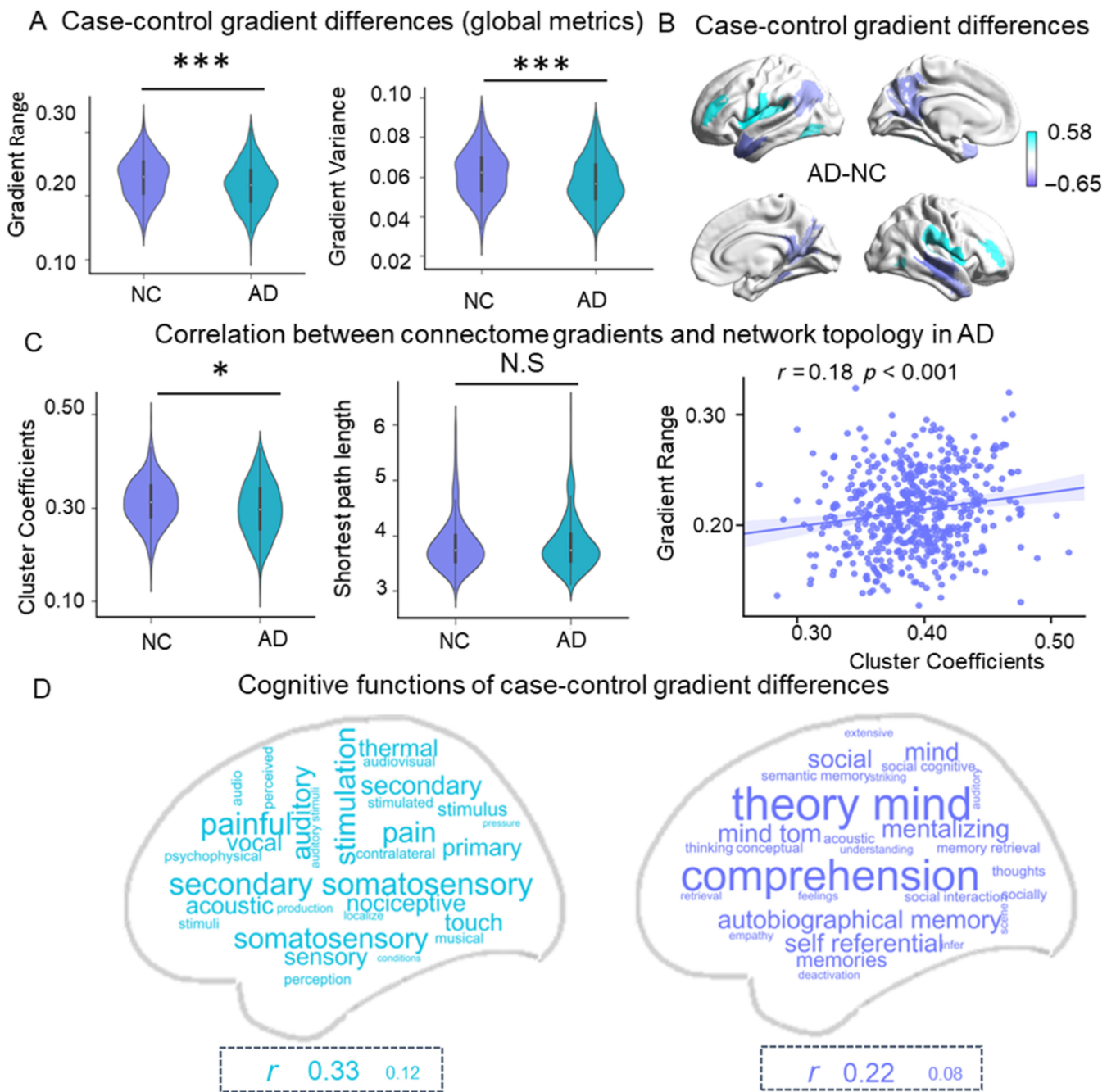


FIGURE 2 | Case-control differences of the gradient metrics. (A) Case-control principal gradient differences in the global metrics (ANOVA analysis). The statistical significance threshold for the global gradient metrics was set to $p < 0.05$. (B) ROI-wise comparisons in the primary-to-transmodal gradient between groups ($p < 0.05/N$, $N = 210$, FWE correction). (C) Case-control differences in the clustering coefficient (C_p) and the association between C_p and gradient range ([ANOVA analysis]). The statistical significance threshold for the network topology metrics was set to $p < 0.05$. (D) Association between meta-analytic cognitive terms and AD-related connectome gradient alterations. Word cloud plots show the relationships between meta-analytic cognitive terms and AD's higher (cyan) or lower (purple) gradient scores. The font size was scaled according to the correlation of meta-analytic cognitive terms and AD-related connectome gradient alterations.

shown in Table S2. In AD, meta-analytic cognitive terms are mainly involved in perceptual processes (e.g., sensory processes regions), which showed increased principal gradient (AD > NCs) (r range, $0.12 < r < 0.33$, Figure 2D). In contrast, the regions with high-order cognitive terms (e.g., social, memory, and theory mind) are associated with the decreased principal gradient (AD < NCs) (r range, $0.08 < r < 0.22$, Figure 2D).

3.4 | Association Between AD-Related Gradient Change and Cognitive Differences

We investigated the relationship between the altered AD gradient with significantly different brain regions and clinical cognitive variables. We performed a Pearson correlation between MMSEs and gradient scores. The relationships between gradient scores

and MMSE scores are shown in Figure S11. The gradient scores of the left anterior superior temporal sulcus (MTG_L_4_4), left ventral dysgranular and granular insula gyrus (INS_L_6_4), and medioventral occipital cortex (MVOcC_5_5) are positively correlated with MMSE scores. The gradient scores of the right precentral gyrus (PrG_R_6_5), right superior temporal gyrus (STG_R_6_2), left superior temporal gyrus (STG_L_6_3), and left lateroventral fusiform gyrus (FuG_L_3_3) are negatively correlated with MMSE scores.

3.5 | Association Between Alterations in Connectome Gradient and Gene Expression Profiles

We assessed the spatial relationship between regional gene expression scores (Figure 3A) and the gradient difference (AD vs. NC) (Figure 3C). The first PLS component (PLS1) accounted for the highest variance (13.2%) of the AD-related alterations in the principal gradient (Figure 3B). In addition, the PLS1 scores are significantly associated with the case-control gradient difference ($r=0.36$, p adjust = 0.01, Figure 3C,D).

We performed a gene set enrichment analysis on the ranked genes to explore the involved biological pathways of the AD-related alterations in the principal gradient (Xia, Liu, et al. 2022). The Gene Ontology (GO) results showed that the PLS1-coded gene proteins enriched in potassium ion transport and protein-containing complex remodeling ($p < 0.05$, FDR corrected, Figure 3E). We further investigated the spatial correlation between the altered functional gradients and PET/SPECT receptor maps. The neurotransmitter system, specifically the receptor 5HT1a-2, has the most significant correlation with the altered functional gradients (Figure S17).

3.6 | Validation Analysis

To examine the reproducibility of abnormal gradient patterns, we used the Willard atlas (region number 433) (Altmann et al. 2015), and Schaefer 2018 atlas (region number 998) (Schaefer et al. 2018) with the same analysis steps. Both brain atlases showed that the principal gradient was organized from the primary visual/sensorimotor networks to the DMN (Figure S12). We mapped the brain regions into surface vertexes and then computed the correlations of functional gradient between different brain atlas at the vertex level. The Pearson correlation between the Brainnetome atlas and the Willard atlas ($r=0.83$, $p < 0.001$), between the Brainnetome atlas and Schaefer 2018 atlas ($r=0.85$, $p < 0.001$) demonstrated that the gradients were replicated with different atlases (Figure S12). In addition, we also explored the effect of the sparsity of the functional connection matrix and motion artifact on the gradient score. Our results were robust to the sparsity of the functional connection matrix and mitigating framewise displacement (all $r > 0.96$, $p < 0.001$, Figures S13 and S14). To assess the impact of age on gradient patterns, we conducted the gradient analysis of both age-matched and unmatched groups. The results demonstrated that the principal gradient patterns are similar in the NC, MCI, and AD groups between age-matched and unmatched groups (all $r > 0.90$, Figures S15 and S16).

4 | Discussion

In this study, we reported connectome gradient dysfunction in AD patients and its association with cognitive terms and transcriptome profile. Functional gradients provide a continuous coordinate system to capture vital characteristics of how

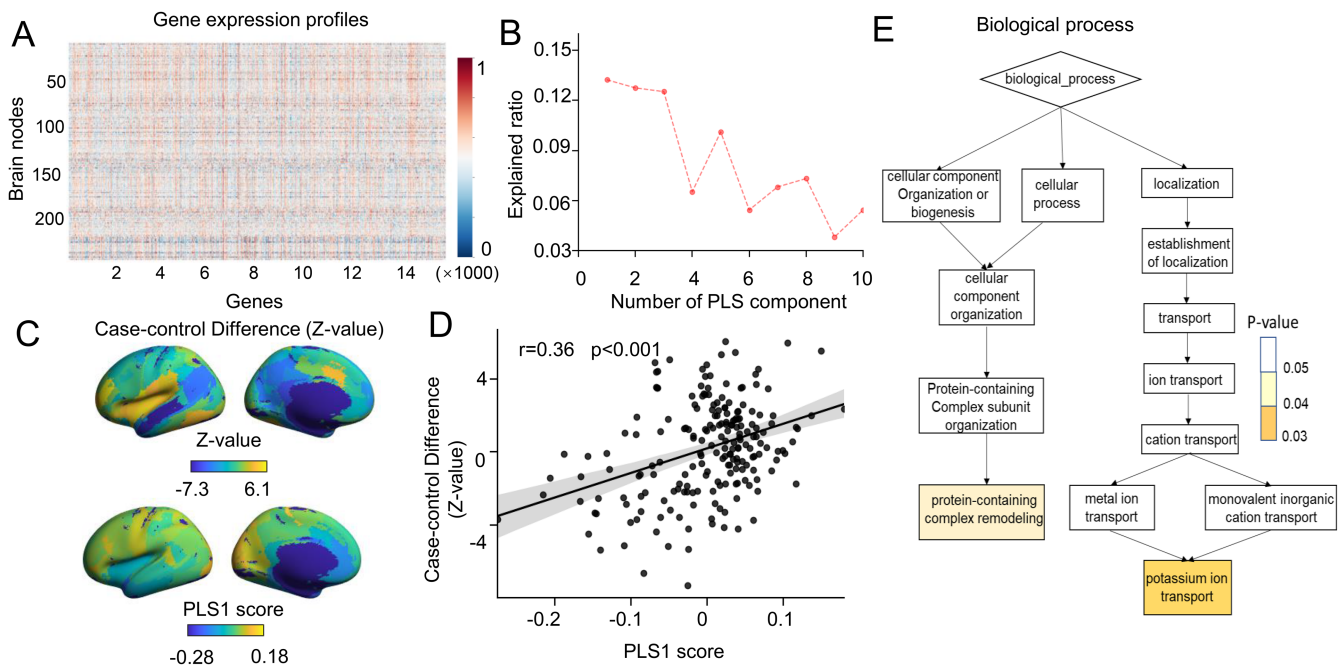


FIGURE 3 | Gene expression profiles related to AD-related alterations in module dynamics and gene expression profiles. (A) Gene expression profiles across 202 brain areas. (B) The explanation ratios for the top 10 components were obtained from the partial least squares (PLS) regression analysis. (C) Spatial patterns in case-control differences and PLS1 scores. (D) The correlation between PLS1 scores and case-control difference. (E) Enrichment analysis of gene ontology terms was observed for PLS1 gene weights.

brain functions are organized. In AD, the principal primary-to-transmodal gradient was suppressed. Our results exhibited global topographic alterations, increased gradient scores in the somatomotor, ventral attention, and frontoparietal regions, and decreased DMN regions. We further used the Neurosynth to assess the meta-analytic cognitive functions in AD. In addition, connectome-transcriptome analysis established a link between AD alterations in connectome gradient and gene expression profiles enriched in potassium ion transport and protein-containing complex remodeling. Together, these results provide new insight into the changes in the macroscale organization of the cortex and the linkage with gene expression profiles in AD.

Macroscale functional gradient techniques at rest by fMRI are of particular interest in detecting intrinsic brain activity. Traditional functional networks are usually defined as discrete entities comprised of brain regions sharing similar features and being strongly connected with each other. Such an approach assumes that there are sharp boundaries between areas representing different functional domains and that connectivity within a given network is homogenous. According to this conventional approach, extensive network abnormalities or impairments in AD patients have been well studied in abundant studies (Ibrahim et al. 2021; Jin, Wang, et al. 2020; Jin, Zhou, et al. 2020; Palop and Mucke 2016). Compared with the traditional functional networks, the macroscale functional gradient techniques provide a continuous coordinate system enabling us to capture additional vital characteristics of how functional domains are organized, extending from unimodal (sensory and motor networks) regions to the transmodal networks (such as DMN). Therefore, the hierarchy alteration has been introduced to explain possible dysfunctions in AD and/or other disorders (Bayrak et al. 2019; Dong, Yao, et al. 2021; Hong et al. 2019).

Functional network hierarchy has been widely regarded as a fundamental organizing principle of the brain. This hierarchy architecture includes unimodal primary and higher-order transmodal networks (Felleman and Van Essen 1991; Mesulam 2012). The brain information flow was guided by hierarchies, transforming unimodal signals into more complex representations (Taylor et al. 2015). Firstly, in line with the study by Hu et al. (2022), the present study showed a gradual axis of connectivity variations that was organized by low-level sensory systems such as the primary and unimodal regions on the one end and the transmodal regions mainly including the DMN on the other end (Dong, Margulies, et al. 2021; Hu et al. 2022; Tian et al. 2020). Meanwhile, the present study focuses on the principal and secondary gradients, which explained 33.17%, 33.06%, and 33.27% of the whole-brain connectome variance in the AD, MCI and NC groups, and found the cortex-wide gradient was altered in AD. Both extremes were contracted relative to the control range, while primary gradients in the central axis increased in AD. The reason might be that the DMN located in the highest hierarchies keeps the maximal spatial distances from the primary system or unimodal regions to restrain input noise or interference and ensure complete information processing from specific stimulation to abstract conceptions (Smallwood et al. 2021; Taylor et al. 2015). Alternatively, in AD patients, the relative capacity to keep the maximal spatial distances from the primary system decreased or impaired, causing a progressive deterioration in cognitive functions and memory. Therefore, the

gradient decreases in the DMN and the middle temporal lobe, reflecting low segregation between the DMN and the remaining networks. These results also suggest a characteristic brain function impairment in AD.

Furthermore, compared with NCs, we found a significant gradient increase in AD in several transmodal frontoparietal, dorsal attention, and sensory regions. The main components of the executive control system are the frontoparietal and dorsal attention networks (Beckmann et al. 2005; Fox et al. 2006; Greicius et al. 2003; Roosendaal et al. 2010; Seeley et al. 2007). Administrative daily control is a complex process including attention control, planning or scheduling complex tasks to accomplish a goal, accessing and manipulating information in long-term memory, and the capacity to monitor current internal and external states (Funahashi 2001). Furthermore, the frontoparietal regions also have a critical role in the memory system (Erberk Ozen and Rezaki 2007). Several neuropsychological and neuroimaging findings show marked impairment of higher cognitive function and relative sparing of sensory and motor function during the progression of AD (Deng et al. 2016; Haxby and Rapoport 1986; Mandal, Joshi, and Saharan 2012). Successfully exercising cognitive functions such as memory, executive function, and motion perception usually depends upon intact sensory perceptual function. Several studies have demonstrated that sensory perception correlates with impaired general cognitive ability (Rizzo et al. 2000). Gradient increase in the sensory network in AD might be a compensatory mechanism of the executive control function and cognitive impairment. This was consistent with our findings that the gradient alterations were significantly correlated with meta-analytic terms involving sensory processes. Hence, AD patients involve a wider range of brain regions, and the increased gradient in these regions might reflect that AD patients need more involvement in executive control cortices. Moreover, in AD patients, the theory of mind is significantly related to the decreased principal gradient. The theory of mind is a comprehensive concept encompassing the understanding of mental states, including beliefs, thoughts, desires, and emotions, which serves the purpose of comprehending, predicting, and evaluating both verbal and behavioral expressions (Wellman, Cross, and Watson 2001) and plays a pivotal role in social and communicative interactions, influencing the comprehension of discourse in both spoken language and written texts. Significant theory of mind and comprehension deficits were presented in AD patients (Lucena et al. 2020; Moreau et al. 2016); AD typically leads to a progressive decline in cognitive functions, including theory of mind, comprehension, and memory. As the disease advances, these cognitive impairments often worsen, making it increasingly challenging for individuals to understand others' mental states, comprehend information, and retain or recall memories.

Our connectome-transcriptome association analysis established a link between AD-related changes in connectome gradients and gene expression enriched in potassium ion transport and protein-containing complex remodeling. AD is a neurodegenerative disease with multiple genetic risk factors (Karch and Goate 2015). Potassium ion transport is crucial for synaptic transmission and plasticity, and changes in potassium channel expression or function could impair synaptic signaling, impacting the strength and efficiency of neuronal connections (Jiang, Yang, and Sun 2020). Furthermore, potassium channels play a crucial role in regulating the resting membrane potential and

action potential firing of neurons. Dysregulation of potassium ion transport can lead to abnormal neuronal excitability, affecting the synchronization and communication between brain regions (de Silva et al. 1998; Etcheberrigaray et al. 1993). The integrity of the potassium ion channel is critical for efficient neurovascular coupling (Longden et al. 2017). A failure of this potassium ion channel-driven neurovascular coupling mechanism has been documented in mouse models of AD (Mughal et al. 2021). Astrocytes, as the interface between neurons and capillaries, also release K⁺ upon detection of neuronal activity. The potassium ion channel impairment rapidly decreases blood flow to the region of reduced neuronal activity (Longden et al. 2017). Recently, one study observed a reduction in pial artery endothelial potassium ion channel function, showing that potassium ion channel function is impaired across the entire cerebrovascular endothelium in AD (Taylor et al. 2015). Studies suggest that in AD, disruptions in ion transport contribute to altered functional connectivity patterns observed (Amadoruge and Barnham 2011; Jia et al. 2021; Kourie 2001; Musunuri et al. 2015). Dysregulated potassium ion transport may result in the breakdown of functional networks that are essential for information processing and integration, which contributes to cognitive decline and the progression of AD-related symptoms.

Previous studies found that hippocampal parvalbumin (PV) interneurons are essential in network oscillations. The hippocampal PV hyperexcitability emerged in the early stages of AD. Restoring the activity of these cells can effectively alleviate cognitive decline in the amyloid precursor protein (APP)/presenilin 1 (PS1) AD mouse model system (Hijazi et al. 2020). The altered expression of genes involved in potassium ion transport may cause the early hyperexcitation of hippocampal PV based on the sequencing data (van Heusden 2023).

Moreover, AD is associated with brain ion transport impairment, which might lead to a cellular ion imbalance. Based on brain samples of 12 NCs and 16 AD, a postmortem found that AD patients are associated with significant increases in potassium ions, suggesting an impaired cell ion homeostasis triggered by A β in AD brain, which could significantly affect brain electrophysiological activity, contributing to the pathophysiology of AD (Vitvitsky et al. 2012). Protein-containing complexes play an essential role in cellular functions. It is widely known that AD is a complex neurodegenerative disorder with impaired protein activities. Apolipoprotein E (APOE) can support lipid transport and injury repair in the brain, and its polymorphic alleles are the primary genetic determinants of late-onset AD risk (Liu et al. 2013). The APOE mapped to protein-containing complex remodeling, whereby high-density and low-density lipoprotein particle remodeling greatly influence AD pathogenesis (Regan et al. 2012). This might explain the gradient alteration-associated gene expression enriched pattern.

There are several limitations in the present study. First, the image acquisition scanning protocols across the sites were different; even though we performed statistical comparisons for each site and combined the results through meta-analysis to attenuate the effects of these confounds. Second, longitudinal neuroimaging and neurobiological data are needed to verify AD-relevant functional alterations in future studies. Third, a

significant proportion of the study participants lacked CSF or PET measures, potentially leading to false positives in diagnosing AD. To address this limitation and enhance the reliability of our findings, we will collect and obtain access to datasets encompassing MRI and CSF measures in the future. Finally, we used the gene expression data from six healthy brains from donors of different ages, ethnicities, and sexes to investigate the association between the Z-map of AD gradients and gene expression, which may have led to variance in gene expression data.

5 | Conclusions

In the present study, we investigated the functional gradients in a biobank, including data from seven neuroimaging centers. The primary-to-transmodal gradient decreases in the DMN regions and increases in several frontoparietal and dorsal attention regions in AD. These gradient alterations were significantly correlated with the perceptual processes, memory, and theory of mind. Additionally, the altered connectome gradient is associated with the gene expression profiles enriched in particular biological processes of potassium ion transport and protein-containing complex remodeling. Collectively, the present study provides a novel framework to characterize neurobiological underpinnings and connectivity alterations in AD.

Author Contributions

Dawei Wang and Zhuangzhuang Li equally contributed to this study as co-first authors. Dawei Wang and Zhuangzhuang Li acquired and interpreted the data, wrote the manuscript, and conducted the analysis. Pindong Chen and Fan Yang completed the data processing. Pindong Chen, Kun Zhao, Hongxiang Yao, Bo Zhou, Jie Lu, Xi Zhang, Ying Han, and Yuqi Chen interpreted the data, edited the first draft of the manuscript, critically reviewed it, and approved the submitted version. Yong Liu and Pan Wang designed the study, acquired and interpreted the data, wrote the manuscript, and approved the submitted version.

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Ethics Statement

Data used in the present study was collected from four hospitals. The Medical Ethics Committee of PLA General Hospital, the Medical Ethics Committee of Qilu Hospital of Shandong University, and the Medical Ethics Committee of Tianjin Huanhu Hospital approved this study. Written informed consent was obtained from each enrolled subject or their authorized guardian. The study was approved by the Medical Research Ethics Committee and Institutional Review Board of Xuanwu Hospital ([ClinicalTrials.gov](https://www.clinicaltrials.gov) identifiers: NCT02353884 and NCT02225964).

Consent

All authors have approved the contents of this manuscript and provided consent for publication.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.